Selective accounting and matched governor comparisons

20/32 trials completed; stopped trials remain in the figures. In the primary TCP block, K4 completed 18 s, reduced net human output by 60.9% relative to K0, and reached 0.148221 W peak human output. Its maximum 10 N force error was 1.647 N and contact absence was 0.00000 s. It achieved 87.3% of K0 measured progress in the same 18 s. K3 stopped at 10.754 s; this supports a completion benefit for the governor on this fixture, not a measured K3 post-release recovery peak.

Both K3 and K4 complete the helping/blocking case. From its 11.7 s release through 18 s, K3/K4 peak x speed is 0.056434/0.064006 m/s and maximum 10 N force error is 0.113279/1.416587 N. K4 reduces post-release reference debt from 18.303 to 13.370 mm and gross human output from 0.123659 to 0.081788 J, but increases those recovery peaks. The reference paper's smoother-recovery claim is therefore not reproduced by this governor adaptation.

These are new closed-loop KUKA comparisons inspired by the previous 2-DOF reference. K0 nominal; K1 whole monitored port; K2 residual only; K3 H/R plus human peak power, governor OFF; K4 identical K3 QP, governor ON. The actual plant and nominal 10 N controller are unchanged. A separate contact model evaluates predicted task work at each actual state. Only its explicit tip/table tangential friction changes (1 versus .20). Seven joint torques and a virtual reference clock adapt the reference architecture; they are not its original two-variable Cartesian controller or spatial-anchor governor.

Mode	Protected accounts	Human peak	Governor
K0	monitored only	none	off
K1	whole W	none	off
K2	residual R	none	off
K3	H and R	.15 W	off
K4	same K3 QP	.15 W	on

This is deterministic simulation evidence, not a mathematical proof of general safety or reproduction of the reference paper's numerical results. Prediction uses true-plant dynamics with current measured force held for 1 ms; the task model alone is mismatched. The terminal human guard assumes force increments no greater than 60 N/s times 1 ms. Finite-difference rows are checked by fresh nonlinear rollouts; stopped cases expose solver, nonlinear acceptance, or local feasibility limits. No verified fallback, hardware execution, continuous-time certificate, or universal task-preservation claim is made. Gross outgoing energy is not bounded by a net-work account. K3/K4 have no Stage 2 reserve/backup or hard contact-quality constraints.

Parameters and audit. H/R/W initial=(.12,.30,.30) J; caps=(.15,.30,.30) J; floors=0; CBF rate=5/s; safe-energy offset=25 microjoules. Four .25 ms physics steps per held 1 ms torque. Central FD=.02 Nm, task weight=1, hard rows and nonlinear acceptance. K4 pauses after an H-triggered intervention and resumes below .02 N measured force, with .1 s clock transitions. Independent raw-array audit: PASS; regression suite: 91 tests, 0 failures, 0 errors. An audit PASS checks implementation consistency; stopped trials remain failures to complete.

Provenance. Source commit `dfc09dfcb402`. Source SHA256: `b98d443a9e4b0acf376dd5ba02f0cb2f 2e0b5a1fca5ac9f5d1bee90c3ecf319b`. Frozen design/configuration, raw data, audit and reproduction commands accompany this PDF.

All primary comparisons

Case	Mode/end s	H net J	min H J	min R J	Peak W	Force err N	Loss s	QPs
A	K0/18.000	-0.00000	0.12	0.3	0.0000	0.05	0.0000	0
A	K1 STOP/12.541	-0.00000	0.12	0.3	0.0000	1.85	0.0000	3229
A	K2/18.000	-0.00000	0.12	0.3	0.0000	0.05	0.0000	0
A	K3/18.000	-0.00000	0.12	0.3	0.0000	0.05	0.0000	0
A	K4/18.000	-0.00000	0.12	0.3	0.0000	0.05	0.0000	0
B	K0/18.000	-0.00000	0.12	0.06704	0.0000	0.05	0.0000	0
B	K1 STOP/12.541	-0.00000	0.12	0.2149	0.0000	1.85	0.0000	3229
B	K2/18.000	-0.00000	0.12	0.06704	0.0000	0.05	0.0000	0
B	K3/18.000	-0.00000	0.12	0.06704	0.0000	0.05	0.0000	0
B	K4/18.000	-0.00000	0.12	0.06704	0.0000	0.05	0.0000	0
C	K0/18.000	0.18743	-0.06743	0.1126	0.2951	0.14	0.0000	0
C	K1 STOP/10.624	0.13225	-0.01225	0.1678	0.2651	3.07	0.0000	214
C	K2/18.000	0.18743	-0.06743	0.1126	0.2951	0.14	0.0000	0
C	K3 STOP/10.754	0.10917	0.01083	0.1908	0.1482	14.75	0.0000	588
C	K4/18.000	0.07321	0.04679	0.2268	0.1482	1.65	0.0000	40

A: exact/no human; B: mismatch/no human; C: TCP block; D: mismatch plus TCP block; E: forearm block; F: help then block; S: starved H. Force error is measured relative to the original 10 N target over the available 8.71 s onward window. Totals from stopped prefixes must not be read as complete-run savings. Negative unprotected account values remain visible.

A_exact_nohuman/K1: stopped at 12.541 s: nonlinear_retries_exhausted.
B_mismatch_nohuman/K1: stopped at 12.541 s: nonlinear_retries_exhausted.
C_blocking/K1: stopped at 10.624 s: solver_failure.
C_blocking/K3: stopped at 10.754 s: nonlinear_retries_exhausted.

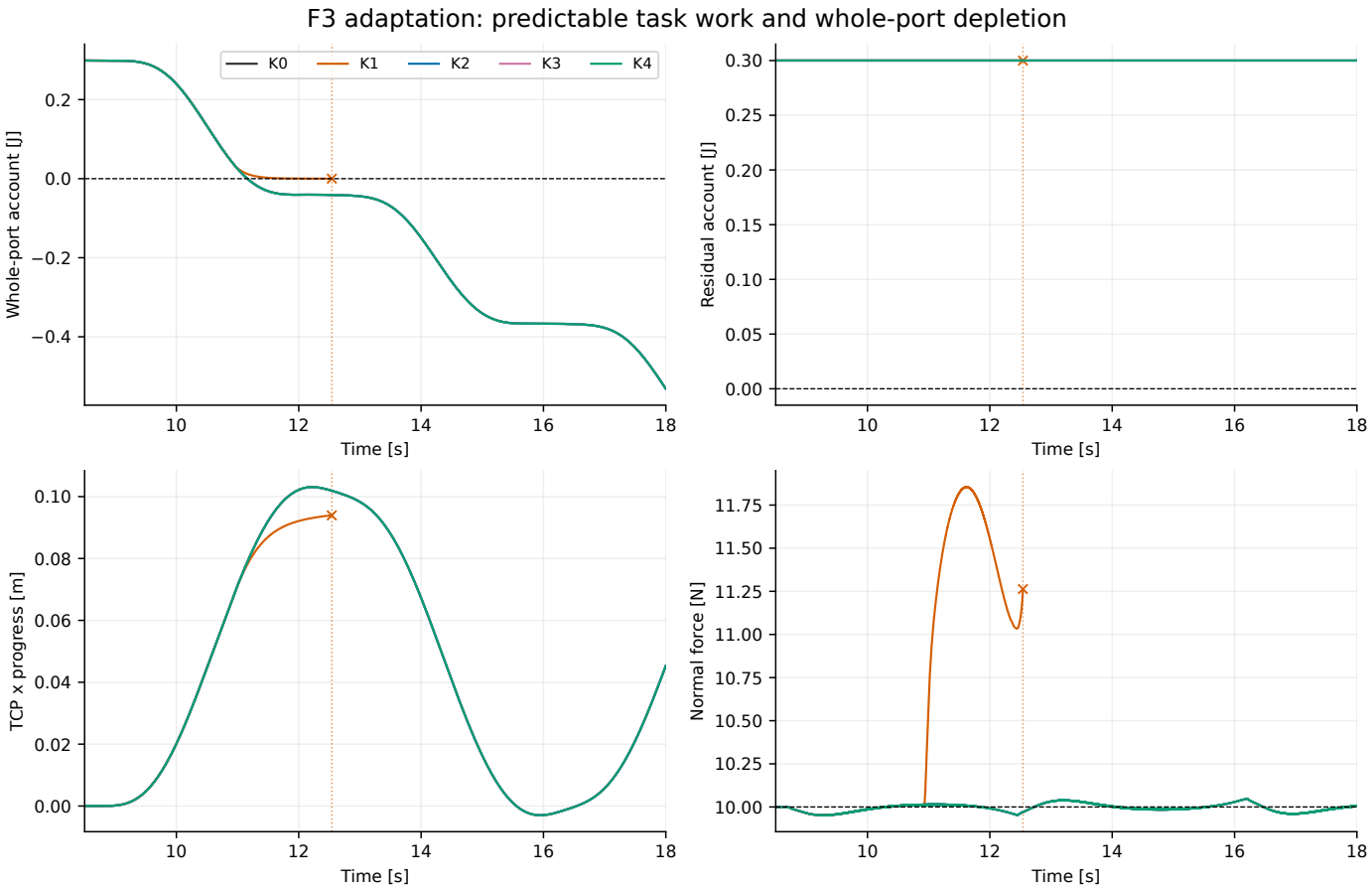
Remaining comparisons and challenge

Case	Mode/end s	H net J	min H J	min R J	Peak W	Force err N	Loss s	QPs
D	K0/18.000	0.18743	-0.06743	-0.2145	0.2951	0.14	0.0000	0
D	K1 STOP/10.624	0.13225	-0.01225	0.1379	0.2651	3.07	0.0000	214
D	K2 STOP/10.752	0.17110	-0.0511	0.07536	0.2951	0.20	0.0000	48
D	K3 STOP/10.754	0.10917	0.01083	0.1839	0.1482	14.75	0.0000	588
D	K4 STOP/14.948	0.07321	0.04679	0.01367	0.1482	1.65	0.0000	153
E	K0/18.000	0.24162	-0.1216	0.05838	0.2389	0.40	0.0000	0
E	K1 STOP/10.335	0.13227	-0.01227	0.1677	0.2442	3.02	0.0000	242
E	K2/18.000	0.24162	-0.1216	0.05838	0.2389	0.40	0.0000	0
E	K3 STOP/10.232	0.08897	0.03103	0.211	0.1482	5.08	0.0000	655
E	K4/18.000	0.03914	0.07605	0.2561	0.1482	0.93	0.0000	10
F	K0/18.000	0.02531	0.01173	0.1617	0.1899	0.25	0.0000	0
F	K1 STOP/13.715	0.02213	0.0149	0.1649	0.1899	3.87	0.0000	3721
F	K2/18.000	0.02531	0.01173	0.1617	0.1899	0.25	0.0000	0
F	K3/18.000	0.01069	0.02634	0.1763	0.1482	0.29	0.0000	383
F	K4/18.000	-0.03118	0.06821	0.2182	0.1482	1.42	0.0000	31
S	K3 STOP/10.363	0.00094	5.566e-05	0.2991	0.0041	17.48	0.0000	1263
S	K4/18.000	-0.06573	0.000683	0.2938	0.0291	11.05	0.0000	60

A: exact/no human; B: mismatch/no human; C: TCP block; D: mismatch plus TCP block; E: forearm block; F: help then block; S: starved H. Force error is measured relative to the original 10 N target over the available 8.71 s onward window. Totals from stopped prefixes must not be read as complete-run savings. Negative unprotected account values remain visible.

D_combined/K1: stopped at 10.624 s: solver_failure.
D_combined/K2: stopped at 10.752 s: nonlinear_retries_exhausted.
D_combined/K3: stopped at 10.754 s: nonlinear_retries_exhausted.
D_combined/K4: stopped at 14.948 s: nonlinear_retries_exhausted.
E_forearm_block/K1: stopped at 10.335 s: solver_failure.
E_forearm_block/K3: stopped at 10.232 s: nonlinear_retries_exhausted.
F_help_block/K1: stopped at 13.715 s: nonlinear_retries_exhausted.
S_starved/K3: stopped at 10.363 s: nonlinear_retries_exhausted.

F3 adaptation: predictable task work and whole-port depletion



Exact model, no human. K2/K3/K4 should be transparent; K1 accounts for ordinary task work. Crosses and dotted lines mark explicit stops, never zero-padded continuations.

K0: COMPLETE, end 18.000 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.3 J; peak H 0.000000 W; measured progress 0.2453 m; QP attempts 0.

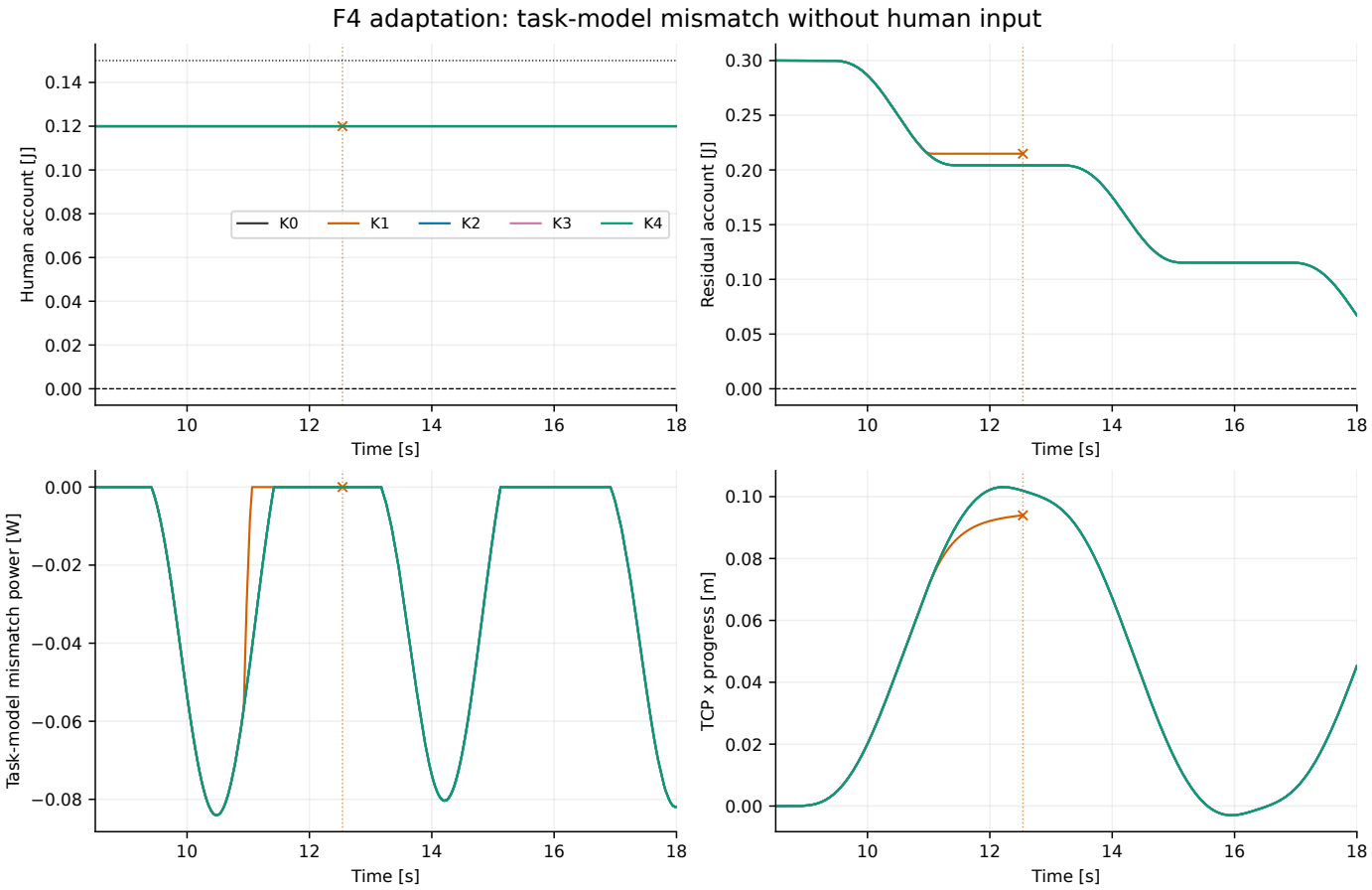
K1: STOP, end 12.541 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.3 J; peak H 0.000000 W; measured progress 0.0939 m; QP attempts 3229.

K2: COMPLETE, end 18.000 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.3 J; peak H 0.000000 W; measured progress 0.2453 m; QP attempts 0.

K3: COMPLETE, end 18.000 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.3 J; peak H 0.000000 W; measured progress 0.2453 m; QP attempts 0.

K4: COMPLETE, end 18.000 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.3 J; peak H 0.000000 W; measured progress 0.2453 m; QP attempts 0.

F4 adaptation: task-model mismatch without human input



Actual friction remains 1; nominal pair friction is .20. Nonzero residual power with zero human power establishes independent accounting. It does not establish robustness to unknown plant dynamics.

K0: COMPLETE, end 18.000 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.0670424 J; peak H 0.000000 W; measured progress 0.2453 m; QP attempts 0.

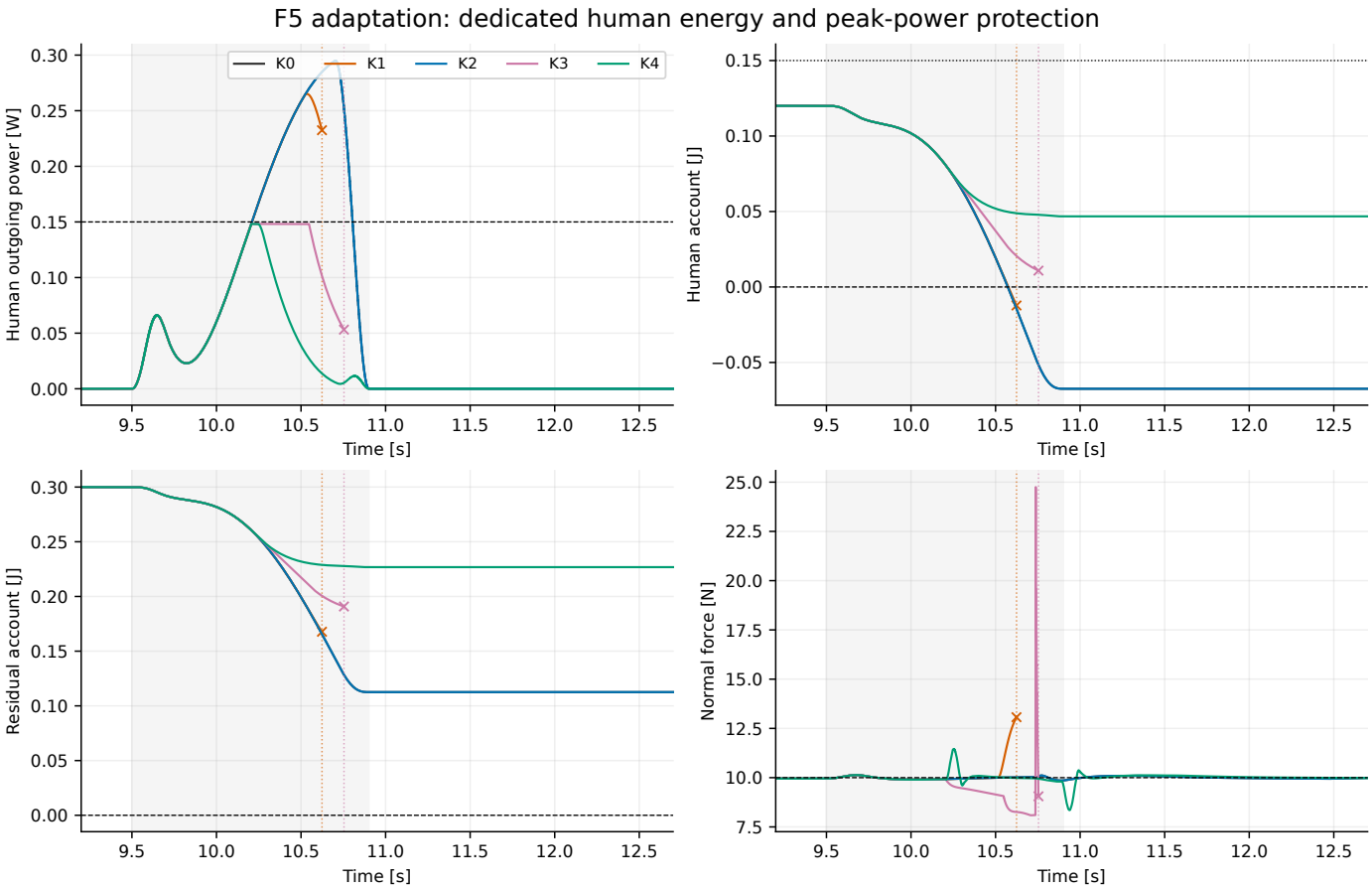
K1: STOP, end 12.541 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.214862 J; peak H 0.000000 W; measured progress 0.0939 m; QP attempts 3229.

K2: COMPLETE, end 18.000 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.0670424 J; peak H 0.000000 W; measured progress 0.2453 m; QP attempts 0.

K3: COMPLETE, end 18.000 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.0670424 J; peak H 0.000000 W; measured progress 0.2453 m; QP attempts 0.

K4: COMPLETE, end 18.000 s; net/gross H output -0.000000/0.000000 J; min H/R 0.12/0.0670424 J; peak H 0.000000 W; measured progress 0.2453 m; QP attempts 0.

F5 adaptation: dedicated human energy and peak-power protection



All modes use the same 5 N TCP block. K3/K4 enforce the .15 W human peak bound and H/R rows. K2 has the larger residual budget and no separate H constraint. Compare completion as well as safe prefixes.

K0: COMPLETE, end 18.000 s; net/gross H output 0.187435/0.187435 J; min H/R -0.067435/0.112565 J; peak H 0.295100 W; measured progress 0.2455 m; QP attempts 0.

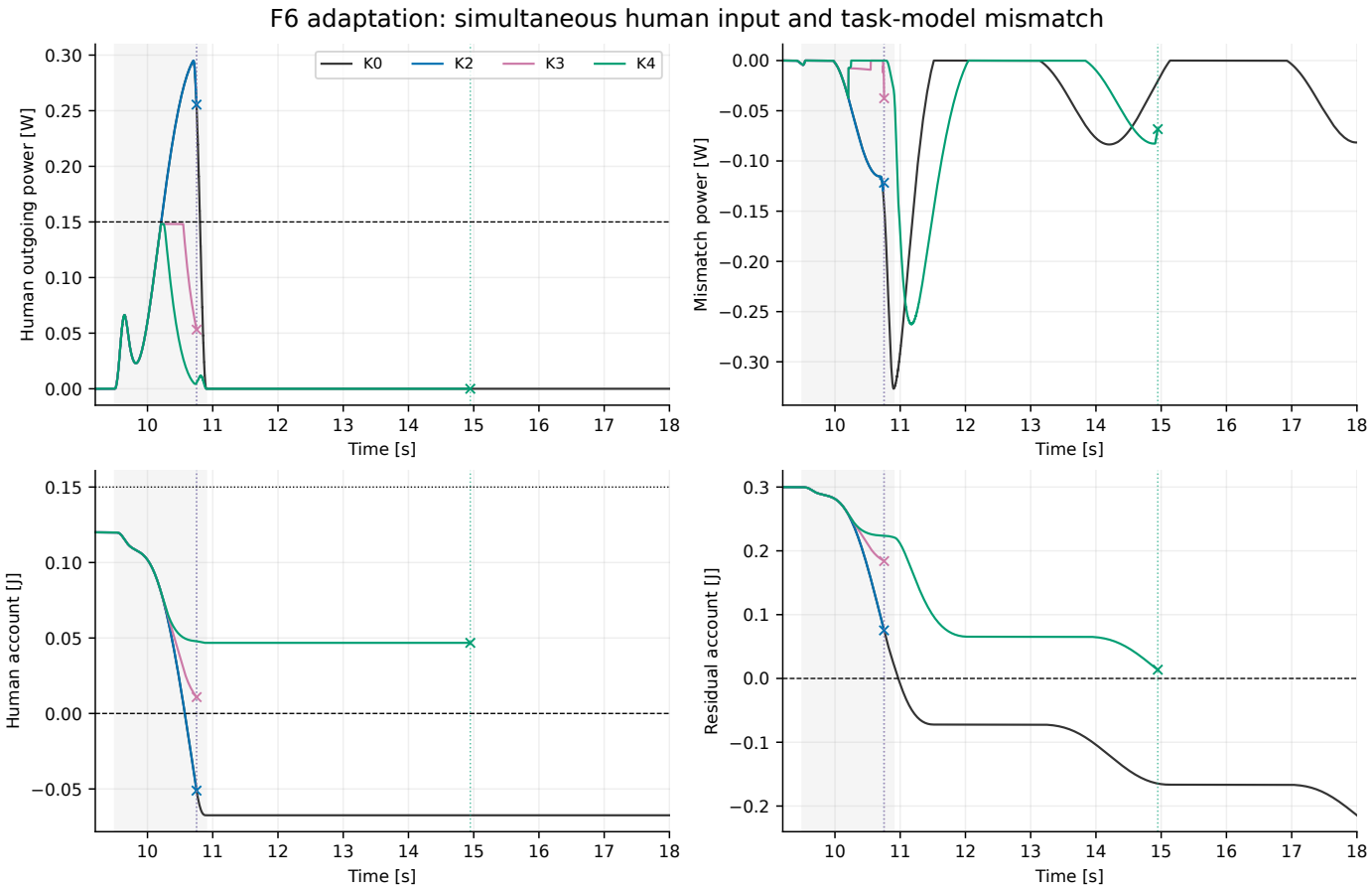
K1: STOP, end 10.624 s; net/gross H output 0.132250/0.132250 J; min H/R -0.01225/0.16775 J; peak H 0.265072 W; measured progress 0.0331 m; QP attempts 214.

K2: COMPLETE, end 18.000 s; net/gross H output 0.187435/0.187435 J; min H/R -0.067435/0.112565 J; peak H 0.295100 W; measured progress 0.2455 m; QP attempts 0.

K3: STOP, end 10.754 s; net/gross H output 0.109175/0.109175 J; min H/R 0.0108255/0.190825 J; peak H 0.148221 W; measured progress 0.0286 m; QP attempts 588.

K4: COMPLETE, end 18.000 s; net/gross H output 0.073209/0.073209 J; min H/R 0.0467912/0.226791 J; peak H 0.148221 W; measured progress 0.2143 m; QP attempts 40.

F6 adaptation: simultaneous human input and task-model mismatch



At every executed sample, pR=pH+(pT-pT_hat). H and R have independently initialized budgets. The cap is applied separately to each account; the raw reconstruction checks both identities.

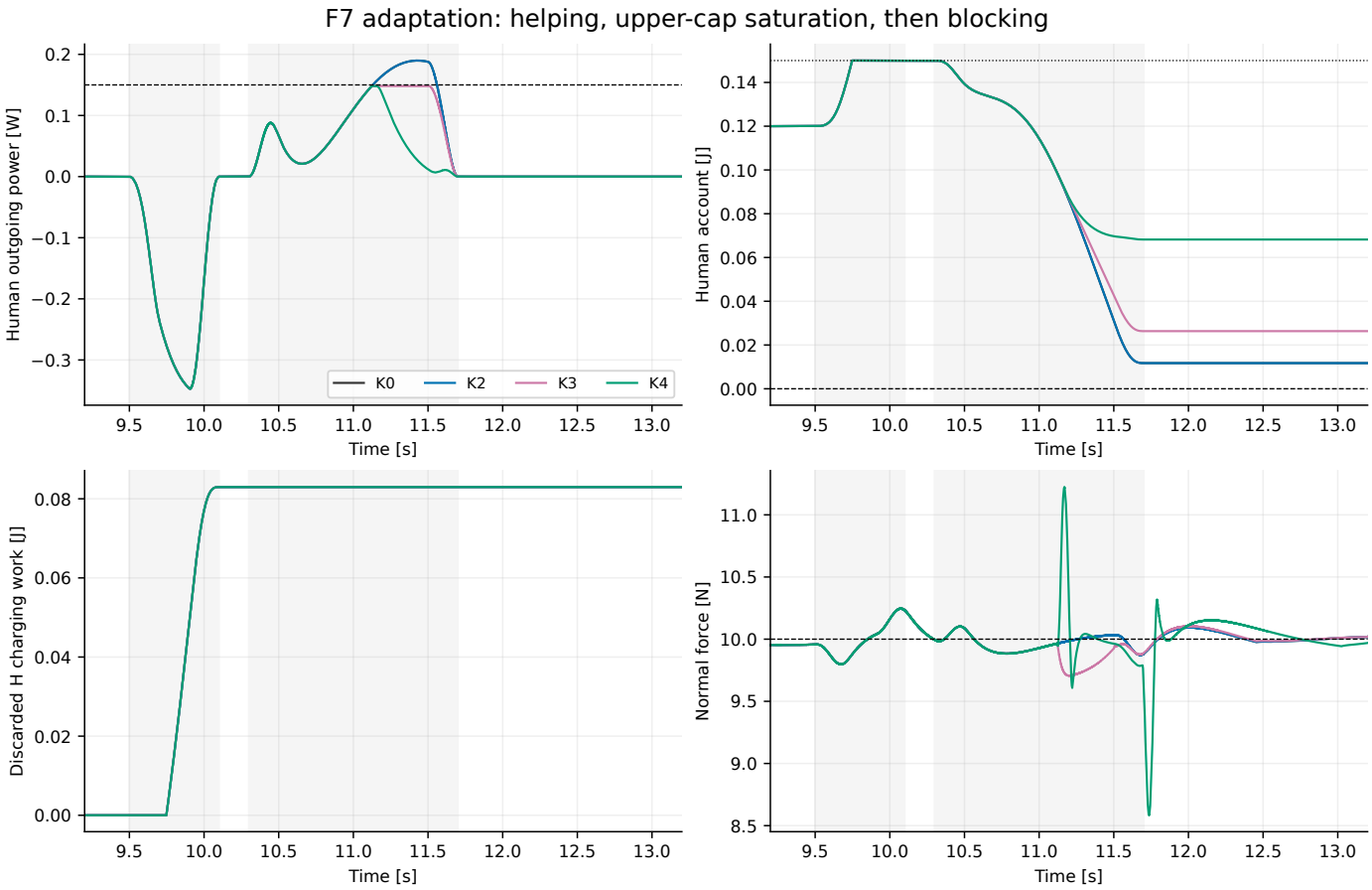
K0: COMPLETE, end 18.000 s; net/gross H output 0.187435/0.187435 J; min H/R -0.067435/-0.214466 J; peak H 0.295100 W; measured progress 0.2455 m; QP attempts 0.

K2: STOP, end 10.752 s; net/gross H output 0.171101/0.171101 J; min H/R -0.0511006/0.0753639 J; peak H 0.295100 W; measured progress 0.0411 m; QP attempts 48.

K3: STOP, end 10.754 s; net/gross H output 0.109175/0.109175 J; min H/R 0.0108255/0.183927 J; peak H 0.148221 W; measured progress 0.0286 m; QP attempts 588.

K4: STOP, end 14.948 s; net/gross H output 0.073209/0.073209 J; min H/R 0.0467912/0.0136669 J; peak H 0.148221 W; measured progress 0.1452 m; QP attempts 153.

F7 adaptation: helping, upper-cap saturation, then blocking



One continuous trial: helping starts at 9.5 s, blocking at 10.3 s. H charges only to .15 J; discarded charging is explicitly logged. Net outgoing work and gross outgoing work are distinct.

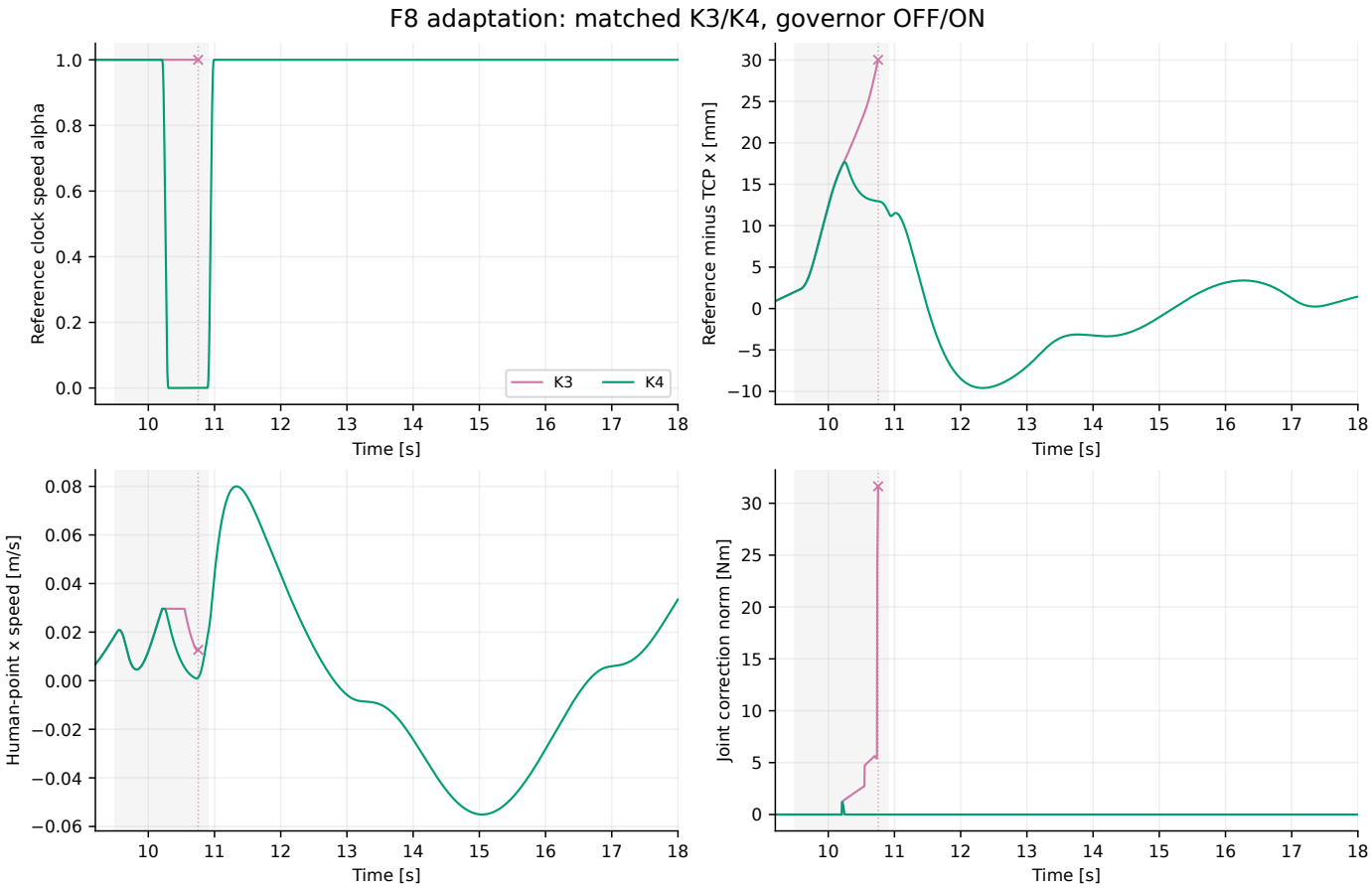
K0: COMPLETE, end 18.000 s; net/gross H output 0.025306/0.138273 J; min H/R 0.0117267/0.161727 J; peak H 0.189865 W; measured progress 0.2456 m; QP attempts 0.

K2: COMPLETE, end 18.000 s; net/gross H output 0.025306/0.138273 J; min H/R 0.0117267/0.161727 J; peak H 0.189865 W; measured progress 0.2456 m; QP attempts 0.

K3: COMPLETE, end 18.000 s; net/gross H output 0.010692/0.123659 J; min H/R 0.0263407/0.176341 J; peak H 0.148221 W; measured progress 0.2455 m; QP attempts 383.

K4: COMPLETE, end 18.000 s; net/gross H output -0.031180/0.081788 J; min H/R 0.068212/0.218212 J; peak H 0.148221 W; measured progress 0.2187 m; QP attempts 31.

F8 adaptation: matched K3/K4, governor OFF/ON

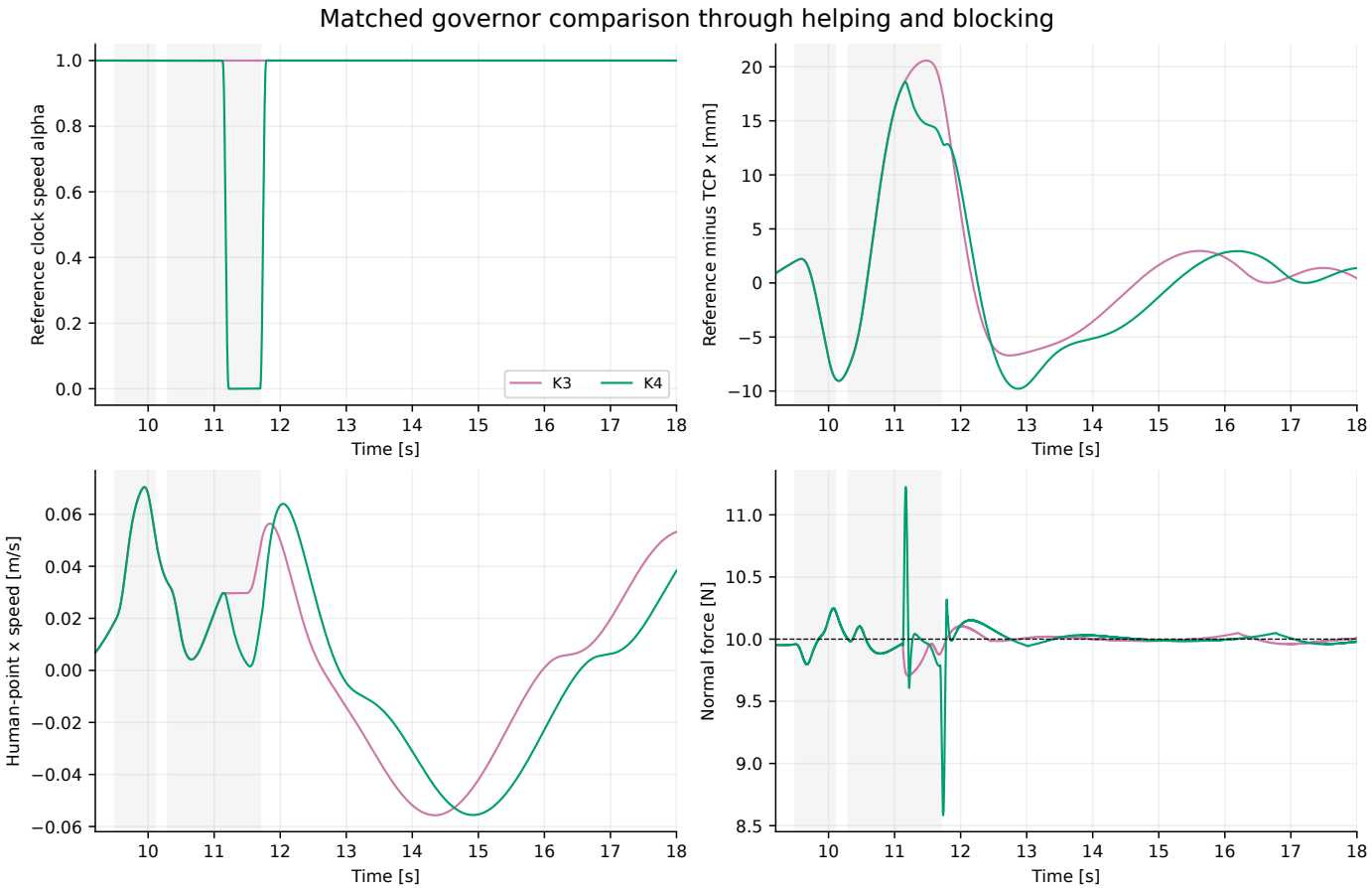


K3/K4 share the QP, bounds, budgets and model; only the reference-clock governor changes. A stopped K3 trace cannot establish a post-release catch-up peak. Joint correction is not a reconstructed Cartesian command.

K3: STOP, end 10.754 s; net/gross H output 0.109175/0.109175 J; min H/R 0.0108255/0.190825 J; peak H 0.148221 W; measured progress 0.0286 m; QP attempts 588.

K4: COMPLETE, end 18.000 s; net/gross H output 0.073209/0.073209 J; min H/R 0.0467912/0.226791 J; peak H 0.148221 W; measured progress 0.2143 m; QP attempts 40.

Matched governor comparison through helping and blocking

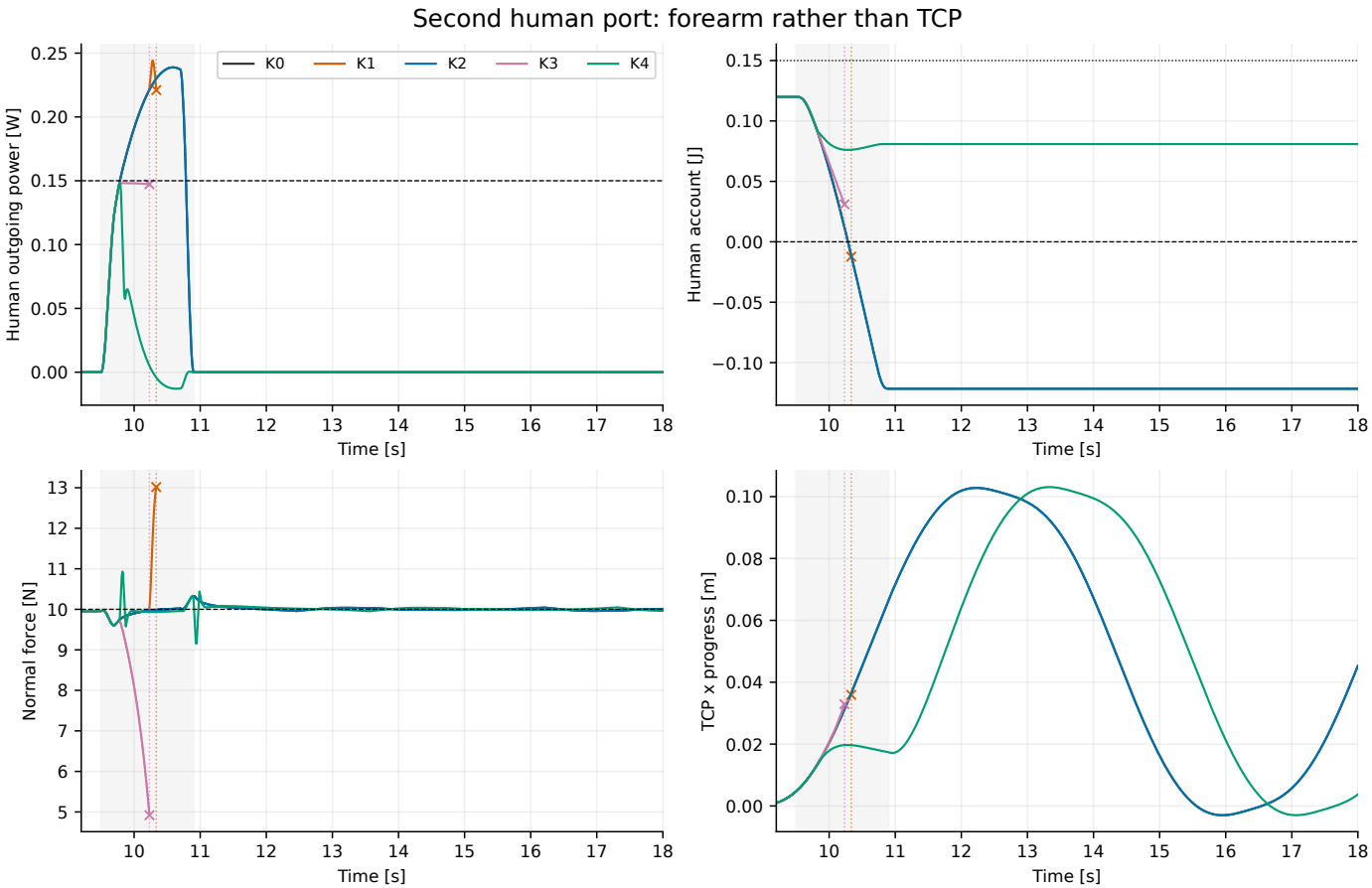


Both trials complete. After the 11.7 s release, K3/K4 peak x speeds are .056434/.064006 m/s and force errors .113279/1.416587 N. The governor reduces human output and reference debt but worsens these recovery peaks.

K3: COMPLETE, end 18.000 s; net/gross H output 0.010692/0.123659 J; min H/R 0.0263407/0.176341 J; peak H 0.148221 W; measured progress 0.2455 m; QP attempts 383.

K4: COMPLETE, end 18.000 s; net/gross H output -0.031180/0.081788 J; min H/R 0.068212/0.218212 J; peak H 0.148221 W; measured progress 0.2187 m; QP attempts 31.

Second human port: forearm rather than TCP



Same world-fixed blocking pulse at link5 local (0,.04,.12) m. Human power uses that actual moving point Jacobian, not TCP velocity.

K0: COMPLETE, end 18.000 s; net/gross H output 0.241623/0.241623 J; min H/R -0.121623/0.0583767 J; peak H 0.238858 W; measured progress 0.2453 m; QP attempts 0.

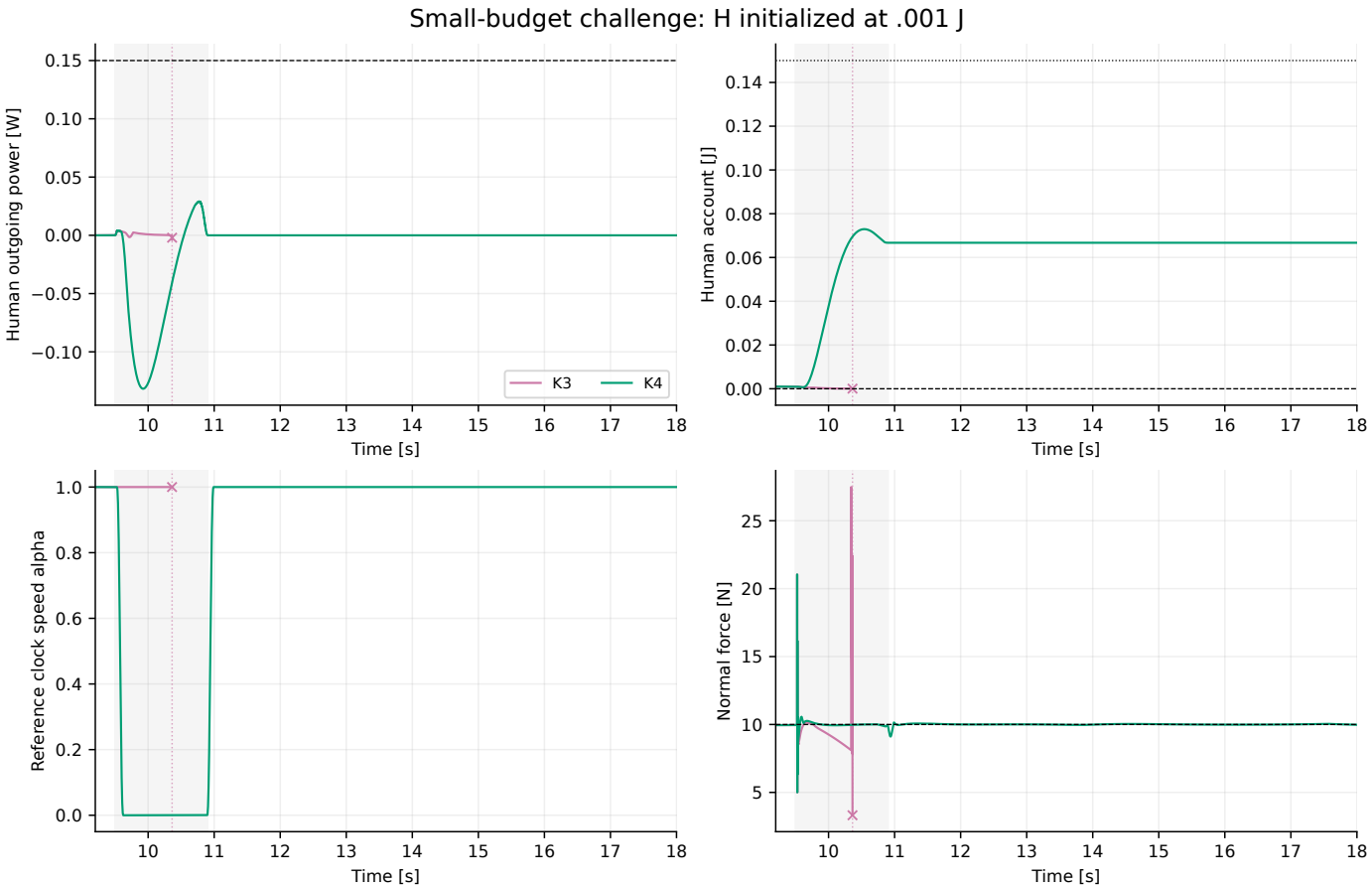
K1: STOP, end 10.335 s; net/gross H output 0.132274/0.132274 J; min H/R -0.0122744/0.167726 J; peak H 0.244181 W; measured progress 0.0359 m; QP attempts 242.

K2: COMPLETE, end 18.000 s; net/gross H output 0.241623/0.241623 J; min H/R -0.121623/0.0583767 J; peak H 0.238858 W; measured progress 0.2453 m; QP attempts 0.

K3: STOP, end 10.232 s; net/gross H output 0.088968/0.088968 J; min H/R 0.0310318/0.211032 J; peak H 0.148201 W; measured progress 0.0329 m; QP attempts 655.

K4: COMPLETE, end 18.000 s; net/gross H output 0.039143/0.043966 J; min H/R 0.076052/0.256052 J; peak H 0.148201 W; measured progress 0.2037 m; QP attempts 10.

Small-budget challenge: H initialized at .001 J



Only the initial H budget changes. Explicit failures are retained with complete pre-action and failure snapshots; no safe-fallback or viability guarantee is inferred.

K3: STOP, end 10.363 s; net/gross H output 0.000939/0.001011 J; min H/R 5.56623e-05/0.299056 J; peak H 0.004108 W; measured progress 0.0058 m; QP attempts 1263.

K4: COMPLETE, end 18.000 s; net/gross H output -0.065725/0.006546 J; min H/R 0.000682986/0.293771 J; peak H 0.029066 W; measured progress 0.2009 m; QP attempts 60.

Which reference claims are supported?

F3: avoid charging predictable task work. K1 first filters at 10.927 s and stops at 12.541 s (nonlinear_retries_exhausted); K2/K3/K4 match nominal bitwise. This is accounting/filter evidence, not proof of unavoidable physical infeasibility.

F4: model error appears separately. No-human mismatch gives maximum |pR-pH|=0.084082 W and minimum R=0.067042 J; H remains .12 J. No residual intervention is required in this finite window.

F5: dedicated human protection. Residual-only K2 reaches H=-0.067435 J and 0.295100 W peak output. K4 completes with H=0.046791 J minimum and peak 0.148221 W. K3 only demonstrates a protected prefix.

F6: simultaneous human and mismatch. Executed power decomposition is independently checked for every mode. Completion and task quality must be read separately from that accounting identity.

F7: bounded bidirectional charging. K4 helping/blocking discards 0.082968 J of H charging at the .15 J cap; final H=0.068212 J. Net H output=-0.031180 J, gross=0.081788 J.

F8: governor changes recovery. K3/K4 have identical pre-governor trajectories and constraints. Primary K3 stops before release. Both K3 and K4 complete the helping/blocking case. From its 11.7 s release through 18 s, K3/K4 peak x speed is 0.056434/0.064006 m/s and maximum 10 N force error is 0.113279/1.416587 N. K4 reduces post-release reference debt from 18.303 to 13.370 mm and gross human output from 0.123659 to 0.081788 J, but increases those recovery peaks. The reference paper's smoother-recovery claim is therefore not reproduced by this governor adaptation.

This is deterministic simulation evidence, not a mathematical proof of general safety or reproduction of the reference paper's numerical results. Prediction uses true-plant dynamics with current measured force held for 1 ms; the task model alone is mismatched. The terminal human guard assumes force increments no greater than 60 N/s times 1 ms. Finite-difference rows are checked by fresh nonlinear rollouts; stopped cases expose solver, nonlinear acceptance, or local feasibility limits. No verified fallback, hardware execution, continuous-time certificate, or universal task-preservation claim is made. Gross outgoing energy is not bounded by a net-work account. K3/K4 have no Stage 2 reserve/backup or hard contact-quality constraints.